



**Tshwane University
of Technology**

We empower people

YEAR: 2023
SEMESTER: 1
ASSESSMENT: Formative Assessment 1

SUBJECT NAME:

Programming Principles

SUBJECT CODE:

PPB115D/PPG115D/PPBF15D/PPGF15D

QUALIFICATION(S):

DPIF20/DPMC20/DPRS20/DPIT20

PAPER DESCRIPTION: CLOSED BOOK **DURATION:** 3 HOURS **PAPER:** ONLY

SPECIAL REQUIREMENTS

- ☐ NONE
- ☐ NON-PROGRAMMABLE POCKET CALCULATOR
- ☐ SCIENTIFIC CALCULATOR
- ☐ COMPUTER ANSWER SHEET
- ☐ GRAPH PAPER
- ☐ DRAWING INSTRUMENTS

OTHER:

INSTRUCTIONS TO CANDIDATES: ANSWER ALL QUESTIONS

CHECK EACH SECTION OR QUESTION FOR ADDITIONAL
INSTRUCTIONS

TOTAL NUMBER OF PAGES INCLUDING COVER PAGE: 11

TOTAL NUMBER OF ANNEXURES: 0

EXAMINER: Miss VM Booï

FULL MARKS: 100

MODERATOR: Mr L. Liebenberg

TOTAL MARKS: 100

STUDENT TOTAL: ____

STUDENT %: ____

INSTRUCTIONS ON HOW TO WRITE THE TEST ON ELECTRONIC CAMPUS (EC):

1. Create a folder on **D:\ Drive** and rename the folder your **student number**.
2. Save all your work in the folder you have created
3. Open a text editor (notepad++ is recommended) or MS Word and save your document in your student number folder.
4. Open the **browser**.
5. Type <http://ec.tut.ac.za> in the address field of the browser and press enter.
6. Click **Login**.
7. Type in student number for user name and if you are using EC for the first time you must use your student number as your password. (Not applicable any more. Please change to the current situation).
8. Alternatively, you must enter your password. (Same here).
9. Click **Login button**.
10. Click [**Continue**]
11. Select [**Student**]
12. Select [**Tests**]
13. Select [**Webtest 4**]
14. Type or Select **PPB115D/PPG115D/PPBF15D/PPGF15D** into the first text field.
15. Type or Select **20232** into the second text field.
16. Type or Select **S** into the third text field (The "**S**" must be a capital letter).
17. Click "**Next**"
18. Click on the "**Write**" button.
19. Click on "**[Start]**" and you will be in the test to submit your answers.
20. Click on [**Section**] on the test to submit each Questions, and click [Go] to submit your answer- Copy your answer and paste it in the space provided.
21. Click [**Test**] to move to the next or previous Questions.
22. Under each question click edit to update your answer.

Question 1
True/False

[10]

Instructions

Log on to EC and proceed to the correct test. Select **Section 1** and provide your answers to this question therein.

Completion time: 10 Minutes

Questions

NO.	STATEMENT
1.1	In a while loop , the loop body always executes at least once.
1.2	A double value cannot be an argument tested in a switch statement.
1.3	The () logical operator is an operator with the lowest precedence out of the three logical operators.
1.4	An index in strings is like a counter variable that indicates a position of a character in the String.
1.5	The subString() method is a method of the String class and it is used to extract part of the string from a String object.
1.6	The difference between int and Integer is that int is a primitive type; Integer is a class.
1.7	All method declarations must always contain the keyword static.
1.8	The methods return type can only be primitive data types and a method can return a maximum of only one value.
1.9	The following is a valid method declaration for a static method named calcPredicate() if the method receives two double arguments. public static double(double a, double b)
1.10	The following method call is equivalent: theFinal =calcFinal(markFA1, markFA2); to calcFinal(markFA1, markFA2);

Question 2
Multiple Choice

[10]

Instructions

Log on to EC and proceed to the correct test. Select **Section 2** and provide your answers to this question therein.

Completion time: 10 Minutes

Questions

- 2.1 The logical structure in which one instruction occurs after another with no branching is a _____.
- a. sequence
 - b. selection
 - c. loop
 - d. case
- 2.2 The loop that performs its conditional check at the bottom of the loop is a _____ loop.
- a. while
 - b. do...while
 - c. for
 - d. All of the above
- 2.3 A loop control variable must always be used three times in these ways:
- a. initialize, display, compare
 - b. initialize, compare, alter
 - c. loop, alter, display
 - d. display, alter, stop
- 2.4 Which statement will correctly increase the value stored in a variable called **amountDue** by 10%?
- a. `amountDue *=1.1`
 - b. `amountDue *=0.1;`
 - c. `amountDue = amountDue * 0.1 + amountDue;`
 - d. `amountDue *=amountDue + 0.1;`
- 2.5 The first position in a String _____.
- a. must be alphabetic
 - b. must be uppercase
 - c. is position zero
 - d. is ignored by the `compareTo()` method
- 2.6 Which of the following is a valid Java statement?
- a. `System.out.println(stringInput.toLowerCase("Value"));`
 - b. `String themePark = new String ("Gold Reef City");`, the value of `themePark.indexOf('R')` is 6.
 - c. `String len= themePark. Length();`
 - d. `If(Character.isUpperCase('C'))`

2.7 The method that determines whether two String objects are equivalent, regardless of case, is _____.

- a. equalsIgnoreCase()
- b. toUpperCase()
- c. equalsNoCase()
- d. equals()

2.8 In Java, methods must include all of the following **except** _____.

- a. a body
- b. a declaration
- c. curly braces
- d. return

2.9 Study the method call below:

quizResults("Status", 9);, which method header is the correct method header for this method call?

- a. public static void quizResults (String stats, int marks);
- b. public static void quizResults (String stats, int marks)
- c. public static void quizResults (status, marks)
- d. public static void quizResults (string stats, int marks)

2.10 Which is a correct call to a method declared as **public static void thisMethod (double price)?**

- a. void thisMethod();
- b. void thisMethod("25.99");
- c. thisMethod(double amount);
- d. thisMethod(1550.23);

Question 3

[17]

Short Questions

Instructions:

Log on to EC and proceed to the correct test. Select **Section 3** and provide your answers to this question therein.

Preparations

- Open Microsoft Word or Notepad and type in your answers therein.
- Question 3.1 to Question 3.3 answer must be typed onto this notepad/word document.
- Submit each question under the slot/section provided on EC.
- Make sure you submit your answers every time when you are done with a question.

Completion time: 20 Minutes

Questions

3.1 Write **only** the required statement/s for the following using **Java syntax** for each.

- 3.1.1 Write a Java while clause that stops the loop when the value in the variable containing the number of students (**noStudents**) is less than the number 0. Only the while clause. (2)
- 3.1.2 Write a **call statement** to a method called **displayCurrentStage**, which will pass the following argument "**Stage 2**". (2)

3.2 Write **only** the required statement or statements.

Consider the following description:

A string variable/object called **stringDetails** has been declared and initialised with the following string:
This is my first Java assessment!

- 3.2.1 Determine the size of the string input called **stringDetails** and store it in an existing variable called **stringLength**. (2)
- 3.2.2 Write a single Java display statement that will display the first three letters of the given string above. (2)
- 3.2.3 Write a single Java statement that will help check for a space in the given string. (2)
- 3.3 Write **only** the required statements using the **switch** structure.
- 3.3.1 Determine the course electives for students' registration using the information in **Table 3.1** below. **Write only the switch statement do not write the whole program.** (7)

Table 3.1

Course	Course Electives
CS CSE	Elective courses: ML, Big Data
IT	Elective courses: Software Engineering
Other	Elective courses: English

Question 4

[23]

Strings and Characters

Instructions

Log on to EC and proceed to the correct test. Select **Section 4** and provide your answers to this question therein.

Preparations

- Create a folder in the **D** drive called “**D:\My Documents\StudentNumber\Question4**”, where **StudentNumber** is your actual student number.
- Create a new file in **Notepad++** called “**ExtractNumbersApp.java**”. Open the file and complete the question as instructed below.

Completion time: 30 Minutes

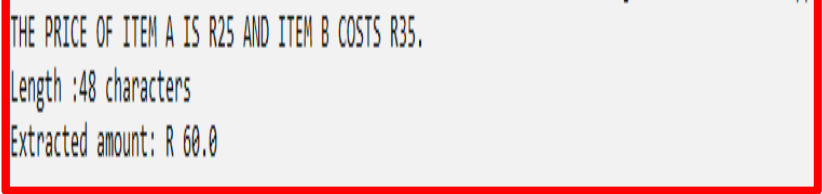
Short Program: Extract Numbers Application

4.1 Develop a Java application that will manipulate a given String. The application must manipulate the String by extracting numeric values from it and calculating the sum of the extracted values. Display the String in upper case, the length of the string and the sum of the extracted numbers.

- Use the following String in your program:

The price of item A is R25 and item B costs R35.

Sample Output:



```
THE PRICE OF ITEM A IS R25 AND ITEM B COSTS R35.  
Length :48 characters  
Extracted amount: R 60.0
```

Figure 4.1

Static MethodsInstructions

Log on to EC and proceed to the correct test. Select **Section 5** and provide your answers to this question therein.

Preparations

- Create a folder in the **D** drive called "**D:\My Documents\StudentNumber\Question5**", where **StudentNumber** is your actual student number.
- Create a new file in **Notepad++** called "**HomeLoanCalculator.java**". Open the file and complete the question as instructed below.

Completion time: 1 hour 30 Minutes

Home Loan Calculator Application

Just Loans is an authorised national credit service provider, where they offer short-term and long-term loans to their clients. They want to expand their business by offering home loans to their clients. They want an application that will help with calculating the monthly repayment amount as well as check if the person qualifies for a home loan. To qualify for a home loan they will look at different factors such as the credit score which must be above the required minimum score of 550 as well as the minimum deposit amount which must not be more than 20%.

Below is the formula that must be used to calculate the monthly payment amount.

$$A = P \frac{r(1+r)^n}{(1+r)^n - 1}$$

Figure 5.1 Formula

Where:

- ***A = payment Amount per period***
- ***P = Principal amount (loan amount)***
- ***r = interest rate per period, which is your monthly interest rate, calculated by dividing your annual interest rate by 12.***
- ***n = total number of payments or periods***

The company wants you to create an application that will help them calculate the home loan repayment amount. The application must allow the company to perform the following tasks:

- **Display the monthly payment amount.**
- **Determine if the client qualifies for a home loan or not.**

- Display if the client qualifies or not.

To successfully create the required application, use the following information:

- **Table 5.1:** UML diagram for the **HomeLoanCalculator** class.
- **Table 5.2:** Description of the **HomeLoanCalculator** class.

Table 5.1

HomeLoanCalculator
<i>+ main(String []): void</i> <i>+ calculateMonthlyPaymnet (double, double, int) : double</i> <i>+ checkLoanQualify(int, int, double) : boolean</i> <i>+ displayMonthlyPayment(double, int): void</i> <i>+ displayQualification(boolean): void</i>

Table 5.2

No.	Method name	Description	Marks
5.1.1	main()	In this method do the following: • Prompt for the principal amount and annual interest rate in decimal (See Figure 5.3). • Randomize the loan term. The loan term ranges from 5 – 25. ○ Call/invoke/use the method calculateMonthlyPayment to calculate the monthly repayment amount. ○ Call/invoke/use the method displayMonthlyPayment to display the monthly payment amount and the term of the loan in years. • Determine if the user needs to check if the client qualifies for a home loan. • If the clients want to check if they qualify, do the following: ○ Prompt the user for the credit score as well as the minimum deposit percentage required (See Figure 5.4- Figure 5.5). ○ Call/invoke/use the method checkLoanQualify to check if the client qualifies for a home loan or not. ○ Call/invoke/use the method displayQualify to display a message whether the client qualifies on not. • This process must continue until the user indicate that he or she does want to check	19

		home loan applications (See Figure 5.2 and Figure 5.6).	
5.1.2	calculateMonthlyPaymnet()	This method will receive three parameters that will consist of the principal amount, the annual interest rate and the loan period. The method must then calculate and return the repayment amount using the formula in Figure 5.1 above.	7
5.1.3	checkLoanQualify()	This method will receive three parameters that will consist of the credit score, minimum credit score and the minimum deposit percentage payable. The method must determine if the clients qualify for a home loan or not. To qualify for a home loan the person must have a good credit score, where the credit score must be the same or above the minimum credit score and must have a minimum deposit percentage which must not be more than 20%. The method must then return true if the client qualifies, and false otherwise.	6
5.1.4	displayMonthlyPaymnet()	This method will receive two parameters that will consist of the monthly payment as well as the loan period. The method must then display loan details as indicated in Figure 5.3 – Figure 5.5.	4
5.1.5	displayQualification()	This method will receive a value indicating whether the person qualify or not. The method must then display the correct qualification message as indicated in Figure 5.4- 5.5.	4

Sample Output

```
C:\Users\p000171\Documents\2023\PPB\PPBF\PPGF\PPBFA1\Questions\java\homeLoanCalculator  
Do you want check home loan applications ? Yes: NO
```

Figure 5.2: Start of the program

```
C:\Users\p000171\Documents\2023\PPB\PPBF\PPGF\PPBFA1\Questions\java\homeLoanCalculator  
Do you want check home loan applications ? Yes: Yes  
Enter the loan amount (principal): R850000  
Enter the annual interest rate (in decimal form): 0.10  
Monthly Payment: R8890.17Period: 16years  
Do you want to check if you qualify for a home loan? Y/y for Yes:
```

Figure 5.3: Prompting and displaying monthly payment

```
Do you want to check if you qualify for a home loan? Y/y for Yes: Y  
Enter your credit score: 520  
Enter the minimum deposit percentage (e.g., 0.20 for 20%): .20  
Sorry, you do not qualify for a home loan based on the given criteria.  
Do you want check home loan applicationsNo ? Yes: Yes  
Enter the loan amount (principal): R850000  
Enter the annual interest rate (in decimal form): 0.12  
Monthly Payment: R9480.78Period: 19years  
Do you want to check if you qualify for a home loan? Y/y for Yes: Y  
Enter your credit score: 650  
Enter the minimum deposit percentage (e.g., 0.20 for 20%): .20  
Congratulations! You qualify for a home loan.  
Do you want check home loan applicationsNo ? Yes:
```

Figure 5.4: Qualify clients

```
Do you want check home loan applications ? Yes: Yes  
Enter the loan amount (principal): R85000  
Enter the annual interest rate (in decimal form): .10  
Monthly Payment: R808.16Period: 21years  
Do you want to check if you qualify for a home loan? Y/y for Yes: Y  
Enter your credit score: 520  
Enter the minimum deposit percentage (e.g., 0.20 for 20%): .20  
Sorry, you do not qualify for a home loan based on the given criteria.  
Do you want check home loan applicationsNo ? Yes:
```

Figure 5.5: Non-Qualifying clients

```
Do you want to check if you qualify for a home loan? Y/y for Yes: Y  
Enter your credit score: 650  
Enter the minimum deposit percentage (e.g., 0.20 for 20%): .20  
Congratulations! You qualify for a home loan.  
Do you want check home loan applicationsNo ? Yes: No
```

Figure 5.6 : No home loan applications and exit

TOTAL FOR SECTION A: 37

TOTAL FOR SECTION B: 63

FULL MARKS: 100